

# AfyaGuide: Responsible AI Statement

Intelligent Health Facility Recommendation System

## 1. People affected and intended use

AfyaGuide is intended to help patients and caregivers find health facilities that are more suitable for a stated healthcare need than a simple “nearest facility” search. It can affect users directly through the facilities it surfaces and ranks, and it can affect health facilities because their recorded services, KEPH level and location influence whether they appear in results. Communities may also be affected where facility information is less complete. The system uses KMHFR facility information and a project-built health-needs taxonomy. It does not diagnose a patient, prescribe treatment or replace clinical judgement.

## 2. Most serious foreseeable harm

The most serious risk is a user treating a recommendation as a guarantee that a facility can provide the required care. If registry information is incomplete or out of date, a person may travel to an unsuitable facility or experience a delay in obtaining care. This risk is more serious for urgent conditions. For that reason, AfyaGuide is positioned as a facility-discovery and recommendation tool, not an emergency, diagnostic or triage service. Results show the factors behind the ranking so that users can make an informed choice and verify important information where necessary.

## 3. Responsibility and cost of errors

The practical cost of an incorrect recommendation is borne primarily by the user or caregiver. The ranking therefore gives the greatest weight to service match and applies a minimum KEPH requirement before distance and facility level are considered. The current scoring weights are design choices, not estimates learned from patient outcomes, and should be treated as assumptions of the prototype. Before any production use, these choices should be reviewed with health-domain experts and tested against labelled recommendation cases and user feedback.

## 4. Redress and accountability

AfyaGuide is a prototype and does not currently make a clinical decision or operate a formal appeals process. A production version should provide a clear way for users and facilities to report an unsuitable recommendation, an incorrect service listing or outdated facility information. Reports should be reviewed by an identified project or health-system owner, with verified corrections incorporated into the underlying data. The system should never prevent a user from choosing another facility or seeking professional medical advice.

## 5. Monitoring and data stewardship

The main monitoring concern is data drift. The cleaned facility dataset contains 12,050 records, while the recommendation taxonomy contains 340 health-need entries. The facility table does not contain a real-time operational-status field, so listed services and facility attributes cannot be treated as a live guarantee of availability. A production system should be reviewed after major KMHFR updates and on a defined schedule for stale service information, missing coordinates, changes in facility capability and low-quality query matches. Serious inconsistencies should trigger correction or temporary removal of affected recommendations.

## Fairness criterion: Individual fairness

AfyaGuide adopts individual fairness as its primary fairness criterion: people with materially similar healthcare needs should receive similarly suitable facility options, regardless of irrelevant personal characteristics. This fits the system because recommendations are based on the requested service, required capability and location rather than protected personal attributes. Geographic equity is also a material concern because differences in data completeness can influence which facilities are eligible for location-based ranking.

## Disaggregated evidence and limitations

Our cleaned facility data contains records from all 47 Kenyan counties, and 98.3% of the 12,050 facility records have both latitude and longitude. The completeness of the underlying information is not uniform across the country. This matters because missing or less detailed facility information can reduce a facility's ability to participate in location-aware recommendations. These findings are evidence of a data-quality risk, not proof of unequal recommendation performance. Before deployment, the team should evaluate recommendation quality separately by county or another meaningful geographic grouping and investigate material differences.

**Deployment position:** AfyaGuide is a transparent facility-discovery prototype. Before production deployment, it should undergo expert validation, formal recommendation evaluation, geographic disaggregation, calibrated match-strength reporting, and a maintained data-refresh and feedback process.